

Zero-Amplitude Continuity Repair, Narrow Posterior Reconstruction, and Independent Reproduction of a High- H_0 Branch in Unified A10 Cosmology

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Abstract

We present a technical follow-up to the A10 cosmology program, focused on the relation between the legacy high- H_0 branch and the later unified implementation. Earlier tests showed an apparent discontinuity: activating the unified A10 switch could strongly distort the CMB fit even in the nominal zero-amplitude limit, making it unclear whether the high- H_0 branch had failed physically or whether the implementation itself lacked the correct legacy limit. We therefore imposed a strict zero-amplitude continuity condition and repaired the unified code so that $V_0 \rightarrow 0$ returns to the intended legacy baseline. After this repair, a finite improvement window in V_0 reappears around the legacy high- H_0 solution, local minimization recovers a fit with $\chi^2_{\text{CMB}} \approx 2853.7$ and negligible SH0ES penalty, and a narrow MCMC run yields a converged posterior centered at $H_0 \approx 72.83$ with $\log_{10} V_0 \approx -10.08$. An independent rerun reproduces the same basin and reaches a best point with $\chi^2_{\text{CMB}} = 2832.86$ at $H_0 = 72.63$. The resulting claim is deliberately narrow: after continuity repair, the unified fixed-trigger A10 model supports a reproducible high- H_0 posterior branch, and the previously observed legacy–unified mismatch is best interpreted primarily as an implementation-level continuity failure rather than as evidence against the existence of a high- H_0 solution.

1 Introduction

The Hubble tension remains one of the central anomalies of precision cosmology: the CMB-inferred Hubble constant in flat Λ CDM differs significantly from direct late-universe determinations such as SH0ES [1, 2]. The A10 cosmology program was developed as an early-time mechanism intended to reduce the sound horizon through a localized energy-transfer episode near $z \approx 5000$, while remaining observationally gentler than simpler constant- ΔN_{eff} -type benchmarks [3, 4, 5]. In its integrated form, the program interprets this localized transfer as a dynamically protected event: a field-space stability window near $\chi \approx 137$ gates the activation of an early-time conversion from a parent component into a stiff daughter sector [9].

The integrated A10 manuscript was framed as a model-class paper rather than a final model-selection paper. It argued that a high- H_0 branch exists in the gated construction, while also emphasizing that stronger likelihood-based comparison would still be required before any broader conclusion could be claimed [9]. The present work addresses a more technical question that emerged during later validation: is the apparent mismatch between the legacy high- H_0 branch and the unified implementation a genuine physical failure, or is it an implementation-level continuity problem?

That distinction is essential. If the unified code does not reduce continuously to the intended legacy limit as the A10 amplitude is taken to zero, then comparisons between the legacy and unified branches are not physically fair. Earlier diagnostics strongly suggested exactly this pathology: in the pre-repair code, the unified switch appeared to activate additional dynamics even when the nominal amplitude vanished. The purpose of the present paper is therefore narrow and technical. We repair the unified implementation so that the $V_0 \rightarrow 0$ limit reproduces the legacy baseline, and we then test whether a repaired high- H_0 branch survives as a posterior support rather than merely as an isolated local minimum.

2 Diagnostic failure mode and repair principle

2.1 Legacy branch versus unified branch

The legacy A10 pipeline had already exhibited a high- H_0 branch near $H_0 \sim 73$, with a self-adjusted matter density near $\omega_{\text{cdm}} \sim 0.1087$ [9]. By contrast, the later unified implementation initially behaved pathologically: toggling the unified switch could dramatically worsen the CMB fit even when the A10 amplitude parameter was nominally set to zero. Under those conditions it was impossible to decide whether the high- H_0 solution had disappeared physically, or whether the unified implementation was simply not connected to the intended zero-amplitude limit.

The central empirical symptom was the lack of continuity in the $V_0 \rightarrow 0$ limit. In the pre-repair unified implementation, switching on the unified branch and setting $V_0 = 0$ did not reproduce the same fit quality as the corresponding legacy baseline. This immediately implied that the unified switch was doing more than activating the intended A10 sector.

2.2 Zero-amplitude continuity as a design requirement

To make the comparison meaningful, we imposed the following requirement:

The unified implementation must reduce continuously to the intended legacy/baseline solution in the limit $V_0 \rightarrow 0$.

In practice, this meant that terms surviving at zero amplitude had to be disabled or explicitly multiplied by an amplitude-dependent gate. Conceptually the repair is simple: the unified branch should not define a different cosmological model in the zero-amplitude limit. Instead, $V_0 = 0$ must correspond to the physically honest statement that the A10 sector is inactive.

2.3 Illustrative continuity audit

Table 1 summarizes representative toggle tests before and after the repair. The key point is not the exact numerical value at each point, but the disappearance of the catastrophic mismatch between the $V_0 = 0$ unified branch and the corresponding legacy baseline.

Table 1: Continuity audit at representative points.

Case	Branch / setting	Pre-repair χ_{CMB}^2	Post-repair χ_{CMB}^2
P0 baseline-like point	unified, $V_0 = 0$	30595.5	3688.06
P1 low- H_0 control point	unified, $V_0 = 0$	42211.7	3688.13
P2 legacy high- H_0 point	unified, $V_0 = 0$	42809.6	3968.77
P0 baseline-like point	legacy / switch off	3688.06	3688.06
P1 low- H_0 control point	legacy / switch off	3688.13	3688.13
P2 legacy high- H_0 point	legacy / switch off	3968.77	3968.77

Diagnostic table only. Its purpose is to show restoration of the intended $V_0 \rightarrow 0$ legacy limit.

3 Repair strategy

The repair focused on two broad classes of terms:

1. **Background-level amplitude-independent modifications.** Some unified background terms modified the total energy density or expansion history even when the nominal A10 amplitude vanished. These terms were either disabled in the zero-amplitude limit or explicitly gated by an amplitude-dependent factor.
2. **Perturbation-level drag/source terms.** Perturbation-level interaction terms depending on an effective drag rate were required to vanish identically when the A10 amplitude vanished. This ensured that turning on the unified switch without finite amplitude did not alter the perturbation system relative to the intended legacy limit.

After the repair, toggle tests showed that the unified branch with $V_0 = 0$ reproduces the legacy/baseline behavior. This restores the minimum requirement for a fair branch comparison.

4 Finite- V_0 window around the legacy high- H_0 branch

4.1 Window scan

Once zero-amplitude continuity was restored, we carried out a controlled scan of the repaired unified branch while holding the legacy high- H_0 point fixed and varying V_0 . A narrow but finite improvement window emerged: as V_0 increased from 0 into the range $\sim 10^{-12}$ to $\sim 10^{-10}$, the CMB fit improved relative to the repaired $V_0 = 0$ baseline. At larger amplitudes, the branch eventually encountered a cliff and the fit degraded or became non-finite.

A refined scan showed that the improvement valley deepens substantially by $V_0 \sim 3 \times 10^{-10}$. Specifically,

$$\chi_{\text{CMB}}^2(V_0 = 0) \approx 3968.77, \quad \chi_{\text{CMB}}^2(V_0 = 3 \times 10^{-10}) \approx 3866.51,$$

before the branch deteriorates near $V_0 \sim 10^{-9}$. The important point is not that this scan is yet globally optimized, but that the repaired unified implementation now exhibits a physically meaningful local improvement direction around the legacy high- H_0 solution.

Table 2: Representative repaired-unified window scan around the legacy high- H_0 seed.

V_0	Status	$-\log \mathcal{L}$	χ_{CMB}^2	χ_{shoes}^2
0	finite	2718.14	3968.77	0.00154
10^{-12}	finite	2717.76	3967.95	0.00154
10^{-11}	finite	2715.09	3962.12	0.00154
10^{-10}	finite	2692.89	3912.69	0.00154
3×10^{-10}	finite	2675.37	3866.51	0.00154
10^{-9}	finite, degraded	2886.66	4252.78	0.00154
10^{-5}	non-finite	$-\infty$	∞	0.00154

4.2 Local minimization

Using the repaired high- H_0 point as a seed, we then carried out a local minimization in a restricted parameter subspace. The resulting local minimum reached

$$-\log \mathcal{P} = 2170.50348,$$

with

$$H_0 = 72.999169, \quad \omega_b = 0.023522, \quad \omega_{\text{cdm}} = 0.108207, \quad n_s = 0.997606, \quad \tau_{\text{reio}} = 0.071288,$$

and

$$\chi_{\text{CMB}}^2 = 2853.737, \quad \chi_{\text{BAO}}^2 = 54.205, \quad \chi_{\text{SN}}^2 = 1430.367, \quad \chi_{\text{shoes}}^2 = 0.00154.$$

The significance of this stage is limited but important: after repair, the unified branch can locally support a high- H_0 solution of the same broad type as the legacy branch.

For clarity, we use $-\log \mathcal{L}$ for fixed-point scan/evaluate stages and $-\log \mathcal{P}$ for quantities reported from local optimization or sampled posterior points.

5 Posterior reconstruction

5.1 First repaired-unified narrow run

We next launched a deliberately narrow Cobaya MCMC centered on the repaired high- H_0 branch using the CLASS backend [6, 7, ?]. The purpose was not global model selection, but rather a posterior-level existence test: can the repaired unified fixed-trigger branch survive as a *converged posterior support*, rather than only as an isolated local minimum?

The chain reached the intended convergence thresholds,

$$R - 1 < 0.05, \quad R_{\text{cl}} - 1 < 0.1,$$

and remained entirely within the high- H_0 basin. The resulting posterior summary is shown in Table 3. The main parameters of interest were

$$H_0 = 72.83 \quad (\text{posterior mean}),$$

with median 72.81 and a 16–84% interval 72.68–72.97, together with

$$\log_{10} V_0 = -10.08 \quad (\text{posterior mean}).$$

The posterior-mean fit measures were

$$\chi_{\text{CMB}}^2 = 2846.66, \quad \chi_{\text{shoes}}^2 = 0.0596.$$

Table 3: Posterior summary for the first repaired-unified narrow run.

Parameter	Mean	Median	16%	84%
H_0	72.8271	72.8131	72.6788	72.9680
$\log_{10} V_0$	-10.0807	-10.0587	-10.3416	-9.86158
χ_{CMB}^2	2846.66	2846.10	2841.30	2852.04
χ_{shoes}^2	0.05963	0.04773	0.00687	0.12063

Posterior-level information only; not a strict likelihood ranking against representative point estimates from other branches.

The best sampled point in this narrow run reached

$$-\log \mathcal{P} = 2113.9226,$$

with

$$H_0 = 72.789873, \quad \log_{10} V_0 = -10.156235, \quad V_0 = 6.9785511 \times 10^{-11},$$

and

$$\chi_{\text{BAO}}^2 = 43.224641, \quad \chi_{\text{CMB}}^2 = 2832.9905, \quad \chi_{\text{SN}}^2 = 1426.9379, \quad \chi_{\text{shoes}}^2 = 0.05784366.$$

5.2 Independent rerun

To test reproducibility, we performed an independent rerun with the same narrow repaired-unified Cobaya setup but shifted reference values. This rerun also satisfied the stopping criteria and remained in the same high- H_0 basin. Its best sampled point was

$$-\log \mathcal{P} = 2112.9453,$$

with

$$H_0 = 72.627912, \quad \log_{10} V_0 = -10.017149, \quad V_0 = 9.6128335 \times 10^{-11},$$

and

$$\chi_{\text{BAO}}^2 = 41.494615, \quad \chi_{\text{CMB}}^2 = 2832.8553, \quad \chi_{\text{SN}}^2 = 1426.3804, \quad \chi_{\text{shoes}}^2 = 0.1570049.$$

This rerun does not merely drift into a vaguely similar region; it reconstructs the same qualitative branch: high H_0 , small SH0ES penalty, and χ_{CMB}^2 in the low 2830s to mid 2840s. We therefore interpret the repaired-unified high- H_0 branch as a reproducible posterior support rather than as an accidental isolated minimum.

Table 4: Convergence diagnostics for the repaired-unified narrow posterior runs.

Run	Samples at stop	Final $R - 1$	Final $R_{\text{cl}} - 1$
First repaired-unified narrow run	173319	0.048085	0.098718
Independent rerun	122242	0.035590	0.084296

6 Comparison with baseline and legacy branches

To avoid mixing posterior summaries with point estimates, we separate branch-level comparison from posterior-level summary. Table 5 compares representative or best sampled points across the main branches, while Table 3 above isolates the posterior-level information specific to the repaired-unified branch.

Table 5: Comparison of representative baseline, legacy, and repaired-unified sampled points.

Model / point	H_0	ω_b	ω_{cdm}	χ^2_{BAO}	χ^2_{CMB}	χ^2_{SN}	χ^2_{shoes}
Baseline Λ CDM best-fit	68.28873	0.02248	0.1192	—	2766.57	1405.31	20.87
Legacy representative (scan_730_unlimit)	72.999169	0.023447069	0.10883147	41.102457	2865.2215	1425.8401	0.00154139
First repaired-unified best sampled point	72.789873	0.023156955	0.10873884	43.224641	2832.9905	1426.9379	0.05784366
Independent rerun best sampled point	72.627912	0.02317632	0.10888077	41.494615	2832.8553	1426.3804	0.1570049

Baseline and legacy rows are representative point estimates. Repaired-unified posterior-level information is reported separately in Table 3.

Several features are noteworthy. First, the baseline Λ CDM fit remains in a low- H_0 basin and pays a large SH0ES penalty. Second, the legacy pipeline shows that a high- H_0 branch exists in principle. Third, after repair, the unified implementation recovers a branch of the same general type but now as a *posterior* rather than merely as a representative scan point. Finally, the independent rerun shows that this repaired-unified branch is reproducible.

7 Interpretation

The main lesson of this paper is narrower than a final claim of model selection, but stronger than a mere existence proof. The repaired-unified A10 branch should not yet be advertised as the globally preferred cosmological model. The present runs are intentionally narrow, the gate position is fixed, and the comparison against broader benchmark classes has not yet been completed. Nevertheless, the repaired branch now satisfies three nontrivial conditions simultaneously:

1. **Continuity.** The unified implementation reduces to the intended legacy limit as $V_0 \rightarrow 0$.
2. **Finite support.** A narrow but genuine finite- V_0 window improves the fit around the high- H_0 seed.
3. **Posterior reproducibility.** The high- H_0 branch survives as a converged narrow posterior and is independently reproduced in a second run.

This changes the interpretation of the earlier failure mode. Before the repair, one could argue that the unified branch simply did not support a high- H_0 solution. After the repair, that reading is no longer compelling. The more natural interpretation is that the earlier discrepancy was dominated by an implementation-level continuity failure.

8 Scope and limitations

Several limitations should remain explicit.

- The present posterior is *narrow* by construction. It demonstrates support for the repaired branch, but not yet global dominance over all cosmological alternatives.
- The trigger position is fixed. We have not yet repeated the repaired analysis in a broad free-gate posterior.
- The unified repair was motivated by numerical and structural diagnostics. Although the repaired behavior is physically more consistent, an even more systematic source-level audit remains worthwhile.
- Full benchmark comparison against other early-time tension-resolution classes remains future work.

These limitations do not erase the result; they define its proper scope.

9 Conclusion

We have presented a technical follow-up to the A10 cosmology program centered on the relation between the legacy high- H_0 branch and the unified implementation. The central advance is the restoration of zero-amplitude continuity: once the unified code is repaired so that $V_0 \rightarrow 0$ returns to the intended legacy limit, the apparent legacy–unified disconnect disappears.

After the repair, the unified implementation develops a finite improvement window around the legacy high- H_0 solution, supports a strong local minimum, and most importantly yields a converged narrow posterior with

$$H_0 \approx 72.8, \quad \log_{10} V_0 \approx -10,$$

together with χ^2_{CMB} in the 2840s and a very small SH0ES penalty. An independent rerun reproduces the same basin and attains a best point comparable to, and slightly improving upon, the first narrow run.

The appropriate interpretation is therefore focused: the repaired-unified fixed-trigger A10 branch is not merely an accidental isolated minimum, but a reproducible high- H_0 posterior branch with independently recovered best sampled points. The next natural step is to enlarge the comparison domain—either by broadening the prior window or by confronting the repaired branch against wider benchmark model classes—while preserving the continuity conditions that proved essential here.

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